A shorter coarse acquisition for the sharp iid Berry–Esseen bound

Fourier estimates, coupled moments, and the remaining count obstruction

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Abstract

We complete the coarse acquisition in the fixed-target proof route. The main improvement feeds a lower bound for the common first moment back into the signed second moment, keeping both constraints attached to one law. Two coarse quadratic passes and four signed passes meet all five numerical targets of the earlier initialization. We independently prove the complete Fourier estimate used by the calculation and supply a distinct exact acceptance contract. A separate harmonic improvement allows conditional span below one; an explicit family of actual iid laws explains why small moment resources cannot replace confinement. The finite base and a complete confinement and count comparison remain necessary. This is not a proof of the sharp conjecture.

1 What has been completed

Let $c = C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$. For a centered, variance-one marginal, write $\rho = \mathbb{E}|X|^3$ and consider the excess

$$a = \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - c\rho.$$

The preceding fixed-target argument gives an attained witness, a closed-support contact inequality, and a conserved identity. At a least failing index $n \ge 42$, sharp predecessors and the external structural estimate give $1 \le \rho \le 7361/4000$ and $|X| \le 96/25$. The earlier two quadratic passes obtain an actual central support gap. These are explicit entry inputs to the present calculation.

The next step used to leave too much freedom in the signed moments. Writing $m = \mathbb{E}X_+$, the defect gives a lower bound for m, while $4m^2 + \beta^2 \le 1$ turns it into an upper bound for $|\beta|$. Applying both before evaluating the resource eliminates the high-rho boxes. Four signed passes suffice. The selected calculation uses 11264 signed boxes and exports all 100 closed probability panels; 30 are nonempty. Exact acceptance also rejects 19 malformed or understated proof objects.

The degree-six phase envelope, six intervening coarse HIGH updates and an unknown extremal-score cap are absent from this selected route. The fixed target c is the coefficient throughout. The resulting component has its own name and hypotheses; its output has not been relabelled as an older certificate with an actual-score premise.

For orientation, the accepted coarse bounds are

$$R_c \le 2.1995174, \quad nD \le 32.20665378, \quad \rho \le 1.490145834, \quad |\beta| \le .50549963, \quad |H| \le 1.83605680.$$

The original coarse contract specifies 239616 signed-box evaluations. The smaller count here concerns that component only, with its small quadratic ancestor counted separately. It is not a runtime comparison or a percentage reduction of the complete proof.

The proof below first establishes the complete Fourier estimate, then explains the signed update and its numerical contract. The final part identifies what a strict count comparison still needs. In particular, the finite prefix through41 is independent of these large-index estimates. The accompanying finite-base audit records its unresolved provider and analytic edges.

2 A uniform Fourier enclosure with explicit input hypotheses

This note independently derives the triangle sixth-order LOW bound and the complete six-part HIGH bound used in the v6 scalar calculation. It also proves their compensated-inversion interface. The conclusions hold for every law satisfying the displayed moment and event hypotheses. Attainment, contact, support acquisition and the acquisition of an event are separate inputs; no numerical acquisition or local-exclusion schedule is accepted here.

For each integer $n \geq N \geq 42$, let $X = X_n$ be a real random variable with $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$. Neither the law nor the real raw threshold $H = H_n$ must be fixed as n varies. Write

$$f(t) = \mathbb{E}e^{itX}, \quad \gamma = \mathbb{E}X^3, \quad e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6), \quad V_n = f^n - e_n.$$

The two complete integrals are

$$I_0(H) = \int_0^{1/2} \frac{|\Im(e^{-itH}V_n(t))|}{t^3}\, dt, \qquad I_\infty = \int_{1/2}^{\infty} \frac{|V_n(t)|}{t^3}\, dt.$$

2.1 The inversion edge, including atoms and moving thresholds

Lemma 2.1. *If a finite signed measure μ has mass and first moment zero and $\int x^2\, d|\mu|(x) < \infty$, then, for every $H \in \mathbb{R}$,*

$$\int (x - H)|x - H|\, d\mu(x) = -\frac{4}{\pi}\int_0^\infty \frac{\Im(e^{-itH}\widehat{\mu}(t))}{t^3}\, dt.$$

The integral on the right is absolutely convergent.

Proof. For real y, substitution and integration by parts give

$$\int_0^\infty \frac{ty - \sin(ty)}{t^3}\, dt = \frac{\pi}{4}y|y|.$$

The boundary terms vanish: near zero the numerator is $O(t^3)$, and at infinity it is $O(t)$. The integrand has the sign of y, since $v - \sin v \geq 0$ for $v \geq 0$. To calculate the constant, nonnegative Tonelli and elementary Laplace integration give

$$\int_0^\infty \frac{1 - \cos t}{t^2}\, dt = \int_0^\infty s\left(\frac{1}{s} - \frac{s}{s^2+1}\right) ds = \frac{\pi}{2}.$$

The absolute double integral of the compensated expression against $d|\mu|(x)$ equals

$$\frac{\pi}{4}\int (x - H)^2 d|\mu|(x) < \infty.$$

Absolute Fubini is therefore valid. The linear compensation disappears because $\int (x - H)d\mu = 0$, proving both assertions. No continuity of a CDF at H is involved. □

Apply this to the sum law minus the scale-$\sqrt{n}$ pushforward of the signed density

$$\phi(x)\{1 + \gamma(x^3 - 3x)/(6\sqrt{n})\}.$$

This signed density has mass one, first moment zero and finite absolute second moment. Its transform after scaling is e_n. Gaussian integration by parts, using $q_z(x) = (x - z)|x - z|$ and $q_z''' = 4\delta_z$, yields

$$\varepsilon_n := \mathbb{E}q_{H/\sqrt{n}}(S_n/\sqrt{n}) - \mathbb{E}q_{H/\sqrt{n}}(Z) - \frac{2\gamma\phi(H/\sqrt{n})}{3\sqrt{n}} = -\frac{4}{\pi n}\int_0^\infty \frac{\Im(e^{-itH}V_n(t))}{t^3}\, dt.$$

Consequently the error $E_n = -n^{3/2}\varepsilon_n/2$ satisfies

$$|E_n| \leq \frac{2\sqrt{n}}{\pi}\{I_0(H) + I_\infty\}. \tag{1}$$

This is an identity and bound for each n, H; it exchanges no limit in n with an integral or a varying law.

2.2 Sixth-order LOW with its actual coefficient bounds

Assume common bounds

$$\mu_4 := \mathbb{E}X^4 \le M \le 111/10, \quad |\gamma| \le g \le 37/20, \quad \mu_6 \le M_6,$$

and put

$$a = (3-\mu_4)/24, \quad z = (10\gamma - \mathbb{E}X^5)/120, \quad |a| \le A, \quad |z| \le Z.$$

Here A, Z, M_6 are nonnegative common ceilings, not quantities inferred from bounded third moments. Define

$$E = M_6/720 + g^2/72 + (1/2 + g^2/144)^3/5, \qquad B = A + Z/2 + E/4,$$

$$a_0 = \frac{N-1}{2N}, \quad b_0 = \frac{45-M}{96}(1-2/N) > 0, \quad \lambda_N = \sqrt{\pi/2}\left(\frac{N}{N-1}\right)^{3/2}.$$

Proposition 2.2. *For all the laws and thresholds above,*

$$I_0(H) \le \frac{L_0 + L_H|H|}{\sqrt{n}} + \frac{L_2}{n},$$

where, for any $c_0 \ge \lambda_N$ and $c_1 \ge \lambda_N/2$,

$$\begin{aligned} L_0 &= c_0 Z + c_1 A g + 251 g^3/17280, \\ L_H &= c_0 A + 251 g^2/4800, \\ L_2 &= E/(2a_0^2) + B^2/(2b_0^3). \end{aligned}$$

Proof. Let $h(t) = \exp(-t^2/2 - i\gamma t^3/6)$. Taylor's integral remainder for e^{itX}, followed by expectation, gives its expansion through degree five with error at most $\mu_6 t^6/720$. For $w = -t^2/2 - i\gamma t^3/6$ and $0 \le t \le 1/2$,

$$|w| \le t^2(1/2 + g^2/144) < 1/6, \quad |e^w - 1 - w - w^2/2| \le |w|^3 e^{|w|}/6 \le |w|^3/5.$$

The first bound follows from $\sqrt{1/4 + y} \le 1/2 + y$; for the last use $e^{1/6} \le (1 - 1/6)^{-1} = 6/5$. Comparing the two polynomials gives

$$f - h = -at^4 - izt^5 + r_6(t), \qquad |r_6(t)| \le Et^6, \qquad |f - h| \le Bt^4. \tag{2}$$

The degree-six term in $w^2/2$ costs $g^2 t^6/72$; it has not been discarded.

Four integrations of $1 - \cos u \ge 0$ show $\cos u \le 1 - u^2/2 + u^4/24$ for every real u. Independence gives $\mathbb{E}(X - X')^4 = 2\mu_4 + 6$, and hence

$$|f(t)|^2 \le 1 - t^2 + (M+3)t^4/12 \le 1 - (45-M)t^2/48.$$

Thus $|f|, |h| \le e^{-(45-M)t^2/96}$ throughout the low interval. Every point on their complex joining segment obeys the same bound. Taylor's integral formula for the polynomial $v \mapsto v^n$ proves

$$f^n - h^n = n(f-h)h^{n-1} + R_2, \quad |R_2| \le \tfrac{1}{2}n(n-1)B^2 t^8 e^{-(n-2)(45-M)t^2/96}.$$

The leading remainder in (2) and this power remainder cost at most L_2/n after division by t^3 and integration: use $n - 1 \ge 2a_0 n$, $(n-2)(45-M)/96 \ge b_0 n$, $n(n-1) \le n^2$, and $\int_0^\infty t^3 e^{-At^2} dt = 1/(2A^2)$, $\int_0^\infty t^5 e^{-At^2} dt = A^{-3}$.

For the remaining leading terms let $b = n - 1$ and $v = b\gamma t^3/6$. Their imaginary part after rotation by Ht is bounded by

$$ne^{-bt^2/2}\{(Z + A|H|)t^5 + bAgt^7/6\}.$$

The Gaussian moments of orders two and four, after dividing by t^3, give $n\sqrt{\pi/2}b^{-3/2}(Z+A|H|+Ag/2)$. This is at most $\lambda_N(Z+A|H|+Ag/2)/\sqrt{n}$. Finally put $u=n\gamma t^3/6$. Direct sine and cosine remainders give

$$|\Im[e^{-itH}(h^n-e_n)]| \le e^{-nt^2/2}(|u|^3/6+|H|tu^2/2).$$

The Gaussian moments of orders six and four respectively give $\sqrt{\pi/2}g^3/(432\sqrt{n})$ times 5, and $\sqrt{\pi/2}g^2|H|/(24\sqrt{n})$. Using $\sqrt{\pi/2}<251/200$ gives precisely the two displayed rational coefficients. Every extended integrand was nonnegative. The only index comparisons were the displayed rational inequalities; in particular the law and H can vary arbitrarily with n. □

For the pinned `sixth` function one has actual support $|X|\le R$ with $1\le R\le 6$ and $\mathbb{E}|X|^5\le E_5$. Then $M_6=RE_5$ and $A=\max(2,M-3)/24$ are valid. Its sharper Z is also licensed: for $Q=X^2-\gamma X-1$ direct multiplication gives

$$\mathbb{E}Q^2=\mu_4-1-\gamma^2,\quad \mathbb{E}X^5=\gamma^3+2\gamma+\mathbb{E}[Q^2(X+2\gamma)].$$

Reflect this auxiliary calculation if necessary so that $t=|\gamma|\ge 0$. For any $r\ge\max(R,2g)$ the two signs of $\mathbb{E}X^5-10\gamma$ are at most

$$P_r(t)=r(M-1)+(10-2M)t-rt^2+t^3,\quad Q_r(t)=r(M-1)+(2M-10)t-rt^2-t^3.$$

The coefficients of μ_4 before replacing it by M are $r-2t, r+2t\ge 0$. If $M\le 5$, $P_r\ge Q_r$ and $P_r\le r(M-1)+(10-2M)t-(r-g)t^2$, whose maximum on $[0,g]$ is at the clipped vertex used in the code. If $M>5$, $P_r\le r(M-1)$ and $Q_r\le r(M-1)+(2M-10)t-rt^2$, again with the stated clipped vertex. For $r=6$, $P_6''=-12+6t<0$ on $[0,g]$. If $M\le 5$ and $P_6'(g)\ge 0$, its maximum is exactly $P_6(g)$; otherwise the preceding quadratic gives $6(M-1)+(10-2M)^2/[4(6-g)]$. For $M>5$ the unrestrained vertex bound is $6(M-1)+(2M-10)^2/24$. Taking the minimum of these proved ceilings and E_5+10g, then dividing by 120, gives exactly the pinned Z. The radius is an actual support input, not a free tuning parameter.

2.3 All six HIGH pieces and their index gates

For independent copies of the same marginal suppose an actual event $\mathcal{E}$ satisfies

$$\mathbb{P}(\mathcal{E})\ge\alpha>0,\quad |X-X'|\ge d>0 \text{ on } \mathcal{E},\quad \mathbb{E}[(X-X')^2\mid\mathcal{E}]\le M_E.$$

Fix $b=9/10$, $0<r\le 1/4$, and

$$h=333/106-(22/7)^3r^2/6>0,\quad \kappa=\alpha(23/48)^2d^2,\quad L(x)=\frac{x+2}{x^2+4x+2},$$

$$S_3=\sum_{j=1}^{20}j^{-3}+1/800,\quad S_5=\sum_{j=1}^{20}(j-r)^{-5}+\frac{1}{4(20-r)^4}.$$

Let $c(u)=u/2-(M+3)u^2/24$ and set

$$c_*=\min\{c(1/4),c(81/100)\},\quad c_q=1/8-(M+3)/384,\quad s=[1-(M+3)53/600]/2.$$

Proposition 2.3. *Suppose $c_* > 0$. For every $n \geq N$, $\sqrt{n} I_\infty$ is at most the sum of the following six terms:*

$$
\begin{aligned}
T_1(n) &= 2\sqrt{n} e^{-nc_*}, \\
T_2(n) &= \frac{M_E}{4\pi^{3/2} h \sqrt{\alpha}} \{S_3 + 3S_5/(n\alpha h^2)\}, \\
T_3(n) &= \frac{M_E \sqrt{n}}{2\pi^2} e^{-n\alpha h^2 r^2}, \\
T_4(n) &= \frac{e^{-n\kappa b^2}}{2b^4 \kappa \sqrt{n}}, \\
T_5(n) &= \frac{\sqrt{n}}{2b^2} e^{-n\alpha(23/48)^2}, \\
T_6(n) &= \sqrt{n} e^{-n/8} \{2L(n/8) + \sqrt{M-1}/3\}.
\end{aligned}
$$

If $s > 0$, T_1 may be replaced by its minimum with $2\sqrt{n} e^{-nc_q} L(ns/4)$. The fourth term may be replaced by its minimum with $\sqrt{n} e^{-n\kappa b^2} L(n\kappa b^2)/(2b^2)$. The common-floor sum at N is valid if

$$
2Nc' > 1 \quad \text{for } c' \in \{c_*, \alpha h^2 r^2, \alpha(23/48)^2, 1/8\}.
$$

For the optional first term also require $2Nc_q > 1$; for the optional fourth term require $N\kappa b^2 \geq 1/4$. If an optional gate fails, the simpler term remains available.

Proof. The triangle inequality separates $|f^n|$ and the entire comparator. On $[1/2, b]$, symmetrization gives $|f|^n \leq e^{-nc(t^2)}$. The function c is concave, so $c(t^2) \geq c_*$ and the integral of t^{-3} is less than two. This gives T_1. Also

$$
c(u) - c_q - s(u - 1/4) = (M+3)(u - 1/4)(81/100 - u)/24 \geq 0.
$$

When $s > 0$ this chord may be integrated to infinity.

For completeness, the tail calculation used here is, for $x = nab^2 > 0$,

$$
\sqrt{n} \int_b^\infty e^{-nat^2} t^{-3} dt \leq \frac{\sqrt{n} e^{-x}}{2b^2} L(x). \tag{3}
$$

Substitution and positive Fubini reduce the integral, without its prefactor, to $\int_0^\infty u(x+u)^{-1} e^{-u} du$. Put $P(u) = u^2 - 4u + 2$, $D = x^2 + 4x + 2 = P(-x)$. Exact polynomial division gives

$$
\frac{1}{x+u} - \frac{-u^3 + (x+8)u^2 - (x^2 + 8x + 20)u + x^3 + 8x^2 + 20x + 16}{D^2} = \frac{P(u)^2}{(x+u)D^2} \geq 0.
$$

Integrate using the factorial moments of e^{-u}; the numerator's integral is $(x+3)D$. Subtracting the resulting lower bound times x from one gives $L(x)$. Applying (3) at endpoint $1/2$ and rate s proves the optional first term.

For $t \geq b$, symmetrization, the actual-event probability bound, and convex Jensen give

$$
|f(t)|^n \leq e^{-n\mathbb{E}\sin^2(t(X-X')/2)} \leq \mathbb{E}[e^{-n\alpha \sin^2(tD/2)} \mid \mathcal{E}], \quad D = |X - X'|.
$$

Positive Tonelli allows a separate partition for every $D > 0$. With $\ell = 2\pi/D$, use the zero cell $[0, \ell/2]$ and nonzero cells $\ell[j - 1/2, j + 1/2]$, $j \geq 1$. In a nonzero cell write $t = \ell(j + v)$. On its core $|v| \leq r$, $|\sin \pi v| \geq h|v|$; on its wings $r \leq |v| \leq 1/2$ it is at least hr. These follow from $\sin x \geq x - x^3/6$ and monotonicity of sin on $[0, \pi/2]$, with directed bounds on π. For $|v| \leq r$ Taylor's integral remainder gives

$$
\{(j-v)^{-3} + (j+v)^{-3}\}/2 \leq j^{-3} + 6(j-r)^{-5} v^2.
$$

Extend the nonnegative Gaussian majorant over the whole real line. Its zeroth and second moments, the Jacobian $t^{-3}dt = D^2(4\pi^2)^{-1}(j+v)^{-3}dv$, and $\mathbb{E}[D^2 \mid \mathcal{E}] \le M_E$ give T_2. The two series tails are bounded by their decreasing integrals after 20. The total wing weight is at most $\int_{\ell/2}^{\infty} t^{-3}dt = D^2/(2\pi^2)$, giving T_3. Extending these sets below b only increases their nonnegative bounds.

In the zero cell put $u = tD/2 \le \pi/2$. Where $u \le 1/2$, $\sin u \ge 23u/24$, and $D \ge d$ yields rate κ. Extending the set to $[b, \infty)$ gives T_4 from $t^{-3} \le t/b^4$; (3) gives its optional improvement. Where $u \ge 1/2$, monotonicity gives $\sin u \ge 23/48$ and the total weight is at most $1/(2b^2)$, proving T_5. Endpoints have zero Lebesgue measure even if the conditional distance distribution has atoms.

Finally $|e_n(t)| \le e^{-nt^2/2}(1 + n|\gamma|t^3/6)$. The first part on the entire $[1/2, \infty)$ uses (3); the second uses $\int_{1/2}^{\infty} e^{-nt^2/2}dt \le 2e^{-n/8}/n$. Pearson's identity $\mu_4 - 1 - \gamma^2 = \mathbb{E}(X^2 - \gamma X - 1)^2 \ge 0$ gives T_6. No comparator tail is dropped.

For floor evaluation, $\sqrt{n}e^{-nc'}$ decreases for $2nc' > 1$, whereas n^{-1} and $n^{-1/2}e^{-nc'}$ decrease for every $c' > 0$. Also $L'(x) = -(x^2 + 4x + 6)/(x^2 + 4x + 2)^2 < 0$. Thus the optional first term decreases under its own gate, and the comparator decreases for $n \ge 5$. The sign of the derivative of $\sqrt{x}e^{-x}L(x)$ is the sign of

$$4 - 10x - 22x^2 - 13x^3 - 2x^4,$$

which is negative at $1/4$ and strictly decreasing on $(0, \infty)$. This proves the optional zero-cell gate. Every comparison is of a common scalar majorant, not of characteristic functions for different laws. □

2.4 The precise closed Fourier conclusion

Under the preceding moment, actual-event and scalar gates, let K_H be any complete HIGH sum evaluated at N, including its permitted optional improvements. Then

$$|E_n| \le \frac{2}{\pi}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\} \le \frac{16}{25}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\}. \tag{4}$$

The second inequality uses $\pi > 25/8$ and nonnegative coefficients. Thus the v6 constant Fourier budget and H-slope have a proved analytic meaning. Complete HIGH bounds for several valid windows may be minimized. The LOW coefficients must remain one complete valid triple.

For the pinned arithmetic implementation, positive denominator roots are rounded downward, numerator roots upward, and $\pi > 333/106$, $\sqrt{\pi} > 4431/2500$, $\pi < 22/7$ are directed correctly. These constants require no numerical transcendental oracle: polynomial division gives

$$22/7 - \pi = \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx > 0,$$

and the tangent addition formula with its first-quadrant branch gives

$$\pi = 16\arctan(1/5) - 4\arctan(1/239).$$

Four alternating terms for the first arctangent and the upper bound $\arctan(1/239) < 1/239$ give $\pi > 333/106 > 25/8$ by rational comparison. Squaring then verifies the two Gaussian constant enclosures used above. Its exponential helper replaces $x \ge 0$ by a downward rational $y \le \min(x, 64)$ and returns an upward rounding of $1/\sum_{j=0}^{64} y^j/j!$. This is at least e^{-x}. The floor argument applies to the underlying analytic bounds, not to a claim that this rounded helper itself preserves strict monotonicity.

This closes only the named Fourier and inversion premises. It does not prove the extremizer/contact reduction, actual support and moment ceilings, any event-acquisition argument, the remaining algebra in the conserved identity, or the final finite/local exclusion predicates.

3 Closing the coarse acquisition by keeping the moments together

The two initial quadratic passes of the preceding note give an actual central support gap. The next calculation keeps the signed moments of one marginal together. It requires no unknown extremal score and uses only the simple quadratic phase price. The point of the refinement is not a finer grid: a constraint already present in the conserved defect can sharply contract a moment box before it is evaluated.

Write $m = \mathbb{E}X_+ = \mathbb{E}X_-$, $\beta = \mathbb{E}[X|X|]$, $\gamma = \mathbb{E}X^3$, $\rho = \mathbb{E}|X|^3$ and $\sigma = \sqrt{10}$. Recall the physical defect $D = (3+\sigma)\rho - \gamma - 3J \geq 0$, where J uses the actual support gaps. The elementary gap inequality $J \leq \rho + 2m$ and signwise Cauchy give

$$D \geq \sigma\rho - \gamma - 6m, \qquad 4m^2 + \beta^2 \leq 1. \tag{5}$$

Thus, on a box with $\rho \geq l$, $\gamma \leq h$ and $D \leq d$, put

$$m_* = \max\{0, (\sigma_- l - h - d)/6\}, \qquad \sigma_- < \sqrt{10}.$$

If $m_* > 1/2$ the box is empty. Otherwise every actual law in it satisfies

$$m \geq m_*, \qquad |\beta| \leq \sqrt{1 - 4m_*^2}. \tag{6}$$

These are simultaneous restrictions, not separately optimized estimates. They narrow the same beta interval before the sign moments and support bounds are recomputed. Each gamma slice starts from its original beta interval; carrying the contracted interval into a different slice would lose coverage.

A second inexpensive constraint comes from a weighted square. Set $\delta = \gamma - \rho\beta$ and $V = \mathbb{E}[|X|^3(|X| - \rho)^2]$. Cauchy with the two factors $|X|^{3/2}(|X| - \rho)$ and $\operatorname{sgn}(X)|X|^{1/2}$ gives $\delta^2 \leq 2mV$. Consequently

$$\frac{\rho\mathbb{E}X^4}{2} - \frac{\mathbb{E}|X|^5}{4} = \frac{\rho^3}{4} - \frac{V}{4} \leq \frac{\rho^3}{4} - \frac{\delta^2}{8m}. \tag{7}$$

Other lower bounds for $V/4$ may be combined by taking their maximum. They cannot be added: each spends the same nonnegative square.

3.1 The precise scalar step

Let $c = C_{\mathrm{E}}$, $c_- \leq c \leq c_+$, and $a \geq 0$ be the fixed-target excess. The raw remainder and nonnegative resource satisfy

$$R_c = \mathcal{U} + n(1 - e^{-H^2/(2n)})\left(c\rho - \frac{\phi(0)D}{6}\right), \qquad \mathcal{U} = (n + H^2)a + \frac{n\phi(0)D}{6} + n^{3/2}\mathcal{L}.$$

Here H is a raw sum threshold, and $\mathcal{L} \geq 0$ is the contact loss. Suppose the incoming bounds are $R_c \leq T < Nc_-l$ and $nD \leq Nd$, with $n \geq N = 42$. They give $D \leq d$ and the density interval

$$\phi(H/\sqrt{n}) \in [\phi_-, 2/5], \qquad \phi_- = (997/2500)(1 - T/(Nc_-l)) > 0.$$

The moment and gap formulas bound the constant charge by A and give $K_1 \in [k_-, k_+]$, $W \in [w_-, w_+]$, where

$$K_1 = \mathbb{E}[X|X|^3] - 3\beta - \tfrac{3}{2}\rho\gamma, \qquad W = J_1 + (\mathbb{E}X^4 - 3)/6.$$

The preceding Fourier proof supplies $|E_n| \le K_F + s_F|H|$. For a formal coefficient $v \in [c_-, c_+]$, define

$$F_v = \max_{k\in\{k_-,k_+\},\ w\in\{w_-,w_+\},\ \varphi\in\{\phi_-,2/5\}} |vk + \varphi w| + s_F,$$
$$Q_v = vl + \min_{\varphi\in\{\phi_-,2/5\}} \varphi\gamma_-/2, \qquad B_v = K_F + vA.$$

Require $Q_v > 0$ at both endpoints. For $h = |H|$ the actual raw remainder is bounded by $B_c + F_c h - Q_c h^2$. For each fixed h, this upper bound is convex in c: F_c is a maximum of absolute affine functions, and the other terms are affine. Therefore it is bounded by the maximum of the two endpoint quadratics, even when A is negative. Set

$$T' = \min\left\{T, \max_{v=c_-,c_+}\left(B_v + \frac{F_v^2}{4Q_v}\right)\right\}.$$

A negative T' excludes the box, since $R_c \ge 0$. Otherwise the previous strict raw gate remains valid. The simple phase lemma gives

$$\eta = \tfrac{1}{2}(c_- l - T'/N) > 0, \qquad \mathcal{U} \le \max_{v=c_-,c_+}\{B_v + F_v h - (Q_v + \eta)h^2\}.$$

A negative maximum of these quadratics excludes the box. In every other box put

$$S = \max_{v=c_-,c_+}\left(B_v + \frac{F_v^2}{4(Q_v+\eta)}\right), \qquad d' = \min\{d, 6S/(N(997/2500))\}.$$

Then $nD \le Nd'$. In particular this is a weighted bound for every $n \ge 42$, not just an estimate evaluated at $n = 42$. A valid phase ceiling is

$$|H| \le \max_{v=c_-,c_+} \frac{F_v + \sqrt{F_v^2 + 4(Q_v+\eta)\max(0, B_v)}}{2(Q_v+\eta)}.$$

The replacement of B_v by its nonnegative part only enlarges this root. All uses of the improved resource follow the raw-license check; there is no circular use of an unlicensed density floor.

3.2 Geometry, support and complete coverage

The actual-gap identity also improves the high-frequency event. If $(-A_0, B_0)$ is the central support gap and $m \le m_u$, the two consequences

$$m(A_0^2 + B_0^2) \ge l - \frac{d}{\sigma_- - 1}, \qquad 6mA_0B_0 \ge (\sigma_- - 1)l - d$$

give

$$(A_0 + B_0)^2 \ge \frac{l - d/(\sigma_- - 1) + \{(\sigma_- - 1)l - d\}/3}{m_u}.$$

The event is the actual opposite-sign pair, not a substituted two-point law. Its probability and conditional distance moment retain their same-marginal derivations. Each high-frequency option includes all six terms and its own damping gates.

For each third-moment cell the support radius can also be reduced before any signed pass. The v5 contact/heat argument tests the boundary polynomial and all six whole-ray derivative signs. We use the first successful radius in $2.50, 2.51, \ldots, 3.84$, with that cell's upper rho and outward beta bound. This implication is independent of the Fourier feedback. The chosen radius then controls every fourth, fifth and sixth moment, negative-gap price and Fourier coefficient in that cell. A proposed radius is never used before its contact gates have passed.

The full signed cover has 24 closed rho cells, eight beta slices, eight gamma slices and two mean slices: at most 3072 nominal boxes per pass. Discarding a slice requires a proved necessary inequality to be inconsistent. In every rho cell the number of active and excluded boxes must equal 128. An excluded rho cell remains recorded in every later pass. All remaining upper bounds are pooled by maxima and lower bounds by minima; this enlarges the admissible set and preserves coverage. The 100 closed probability panels cover $[0, 1]$, and a box meeting a shared endpoint is included in both incident panels.

The exact implementation and independent reviews specify the other moment inequalities: signwise Cauchy, the common first moment, the fourth moment projection, bounded-support moment matrices, and the central and outer actual-gap charges. The source locations and the precise scope of those reviews are listed in the accompanying audit. These are analytic premises of the enclosure; an arithmetic receipt alone does not prove them.

3.3 The resulting conditional entry theorem

Theorem 3.1. *Let $n \geq 42$ and let an attained fixed-target witness have $a \geq 0$. Assume its closed-support contact and conservation identities, the sharp predecessor bound used in the v5 support argument, and the stated external structural bound. Use the actual-law moment and gap inequalities reviewed in this note and the complete Fourier estimates of Section 2, with all their domain gates. Then the two initial quadratic passes followed by the four signed passes verified by* `INITIAL42_FIXED_TARGET_COARSE_V7` *give*

$$R_c \leq 2.1995174 < 2.24, \qquad nD \leq 32.20665378, \qquad D \leq .76682509 < .784,$$
$$\rho \leq 71527/48000 < 1.526, \qquad |\beta| \leq .50549963 < .523, \qquad |H| \leq 1.83605680 < 1.947.$$

There is a complete closed 100-panel probability cover carrying these bounds and the simultaneous noise bounds. Thirty panels are nonempty. The fixed coefficient is enclosed by $.4097321837023 < c < .4097321837025$.

Proof. The two-quadratic ancestor obtains the actual gap on all 24 rho cells. Each signed step preserves the inequalities by the preceding scalar argument and the stated moment/geometry bounds. All radii and Fourier gates are checked before use. Complete incidence accounting and outward pooling preserve every possible witness. The distinct exact acceptance contract replays the ancestor, the support gates and all 11264 signed boxes, and checks the stated targets on every nonempty probability panel. Its proof object and saved fresh receipt are supplied with the source. The inequality for nD is retained at each step before replacing n by its lower bound for an unweighted moment constraint. □

This replaces the coarse initialization in a new fixed-target composition. It does not assert that its output may be relabelled as a legacy certificate with a different actual-score hypothesis. The subsequent confinement and count interfaces need their own transport and complete checks. The degree-six phase envelope and the intervening six coarse HIGH refinements are absent from the selected ancestry.

4 What a strict count comparison actually needs

For a centered variance-one law put $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $m = \mathbb{E}X_+$, $\ell = m/(pq)$, and $X = \ell(B + U - p)$, where $B = \mathbf{1}_{X>0}$ and $\mathbb{E}(U \mid B) = 0$. Write $s = \mathbb{E}U^2$ and $\lambda = ns$. Let $np + H/\ell = k + u$, with an interior integer count k and $|u| \leq 1/2$. All variables are constructed from the same marginal.

For fixed target $c = C_{\mathrm{E}}$, define $a = \sqrt{n}\{F_n(H) - \Phi(H/\sqrt{n})\} - c\rho$. The exact count identity has left side

$$a + \rho_B(c - C_B),$$

where C_B is the actual matched-binomial score. If $a \geq 0$ and the external Bernoulli theorem gives $C_B \leq c$, this is nonnegative. Thus a whole-count upper bound of the form

$$\sqrt{n}\pi_k \lambda B(\lambda) + \frac{\lambda}{n} I(\lambda)$$

is contradictory for $\lambda > 0$ whenever valid constants satisfy

$$B(\lambda) < 0, \qquad m_0 B(\lambda) + \frac{\max(0, I(\lambda))}{N} < 0, \quad \sqrt{n}\pi_k \geq m_0 > 0, \quad n \geq N. \tag{8}$$

The first sign gate is necessary before applying a lower mass bound. Zero noise is handled separately by the Bernoulli theorem.

Neither $D < 0.784$ nor a raw Fourier bound below 2.24 occurs in (8). Those are prescribed outputs of the retained initial acceptor. They supply later acquisition inputs through its particular dependency chain; they are not necessary scalar constants in every possible count proof. Conversely, obtaining a smaller D alone does not establish the conditional support, amplitude, marked-crossing and Gaussian-repair hypotheses needed to produce B, I. The example below shows why.

There is a useful exact test for the remaining marked parameter. On a fixed positive noise value, the retained whole-count formula has the form

$$B(\lambda) = z + \frac{A_\lambda}{z} - F_\lambda, \quad A_\lambda \geq 0, \quad 0 < z \leq 1/U,$$

where $\eta = z\lambda \leq 1$ on the noise band $[L, U]$. Write $F = F_\lambda - \max(0, I(\lambda))/(Nm_0)$ and $Z = 1/U$. Then (8) follows at this noise value precisely when $z + A_\lambda/z < F$ for an allowed z. For $A_\lambda > 0$ the minimum is

$$\min_{0<z\leq Z} \left(z + \frac{A_\lambda}{z} \right) = \begin{cases} 2\sqrt{A_\lambda}, & Z^2 \geq A_\lambda, \\ Z + A_\lambda/Z, & Z^2 < A_\lambda. \end{cases} \tag{9}$$

This follows by differentiating, or completing $(\sqrt{z} - \sqrt{A_\lambda/z})^2$ and checking the boundary. If $A_\lambda = 0$, feasibility is exactly $F > 0$; the infimum zero is not attained. In the interior case the exact rational gate is $F > 0$ and $F^2 > 4A_\lambda$. Equality does not suffice. For a full closed noise band a common z must still satisfy both endpoint gates before using the existing convexity proof; optimizing z separately at the endpoints does not establish that common witness. In particular a zero fractional-part gain makes this whole-count formula nonnegative for every $z > 0$, even if the mark cost vanishes. It cannot be repaired simply by shrinking the weighted resource.

5 The harmonic argument works up to span one

For a real number x, write $\{x\} = x - \lfloor x \rfloor$.

Lemma 5.1 (Extended conditional-span domain)**.** *Let $Z_1, \ldots, Z_j$ be independent real variables, each supported in an interval of length at most b, where $0 < b < 1$. Put $W = \sum_i Z_i$ and $t = \sum_i \operatorname{Var}(Z_i)$. Suppose $K > 0$ satisfies*

$$1 - \cos(2\pi x) \leq Kx \quad (0 \leq x \leq 1).$$

For every $w \in \mathbb{R}$ and every $0 < r \leq 2\sin^2(\pi b)/b^2$,

$$\mathbb{E}\{w - W\} \geq K^{-1}(1 - e^{-rt}). \tag{10}$$

In particular $K = 8$ is valid without a finite scalar certificate. The inherited value $K = 23/5$, or its accepted improvement, may be retained when its separate scalar majorization has been proved.

Proof. Let Z_i' be an independent copy. Since $|Z_i - Z_i'| \le b$ and sine is concave on $[0, \pi]$, the chord from 0 to πb gives

$$\sin^2(\pi(Z_i - Z_i')) \ge \frac{\sin^2(\pi b)}{b^2}(Z_i - Z_i')^2.$$

Consequently, writing $v_i = \mathrm{Var}(Z_i)$,

$$|\mathbb{E}e^{2\pi i Z_i}|^2 = 1 - 2\mathbb{E}\sin^2(\pi(Z_i - Z_i')) \le 1 - \frac{4\sin^2(\pi b)}{b^2} v_i \le e^{-2rv_i}.$$

The intermediate right side is nonnegative because it bounds the nonnegative left side. Taking square roots and multiplying, using independence, gives $|\mathbb{E}e^{2\pi i W}| \le e^{-rt}$. Periodicity then gives

$$K\mathbb{E}\{w - W\} \ge 1 - \Re\mathbb{E}e^{2\pi i(w-W)} \ge 1 - |\mathbb{E}e^{2\pi i W}| \ge 1 - e^{-rt}.$$

No centering or continuous distribution assumption was used. To verify $K = 8$, the function $y - 1 + \cos y$ is nondecreasing for $y \ge 0$, since its derivative is $1 - \sin y \ge 0$. Thus $1 - \cos(2\pi x) \le 2\pi x < 8x$ for $x > 0$; $\pi < 4$ also follows directly from $\pi = 4\int_0^1 (1 + y^2)^{-1}\, dy < 4$. Equality at $x = 0$ is harmless. □

The earlier count proof restricts $b < 1/2$ at this step. The same proof in fact gives Lemma 5.1 throughout $b < 1$. For example, $b = 3/4$ permits the exact rate $r = 16/9$. This extends only the harmonic component. It does not extend other denominator, binomial, support or noise gates. In the retained whole-count notation a sufficient acquired span is $b = A + d_0$; it must be a bound on each actual conditional residual, not on its variance.

For example, on a phase interval $[u_l, u_h]$, the retained individual first-mark denominators are positive when $1 - 2d_0 - u_h > 0$ and $1 - A + u_l > 0$, with $c = 1 - d_0$. On the full phase interval $[-1/2, 1/2]$, these become $d_0 < 1/4$ and $A < 1/2$. They imply $A + d_0 < 3/4 < 1$, so the extended harmonic lemma adds no further span restriction there. The optional common-mark bound has additional gates. None of these inequalities alone establishes the final score, and a larger span gives a smaller harmonic rate that must be used in that score.

Proposition 5.2 (The boundary is a real obstruction)**.** *No strictly positive uniform rate in* (10) *is possible for all independent positive-variance summands of support span at most one. Even a uniformly positive rate as $b \uparrow 1$ is impossible.*

Proof. For $0 < \varepsilon < 1$, let Z equal $\varepsilon - 1$ with probability ε, and ε otherwise. It is centered, has positive variance $\varepsilon(1 - \varepsilon)$ and span one. For j independent copies, $W = j\varepsilon - M$ with integer-valued M, so $\{j\varepsilon - W\} = 0$ identically. The right side of (10) is positive for every $r > 0$ and finite $K > 0$.

For the boundary limit, the centered variable taking values $\pm b/2$ with equal probabilities has variance $b^2/4$ and characteristic modulus $|\cos(\pi b)|$. As $b \uparrow 1$, this modulus tends to one while the variance tends to $1/4$. A positive uniform exponential decay rate would contradict these limits. Directly for (10), choose $w = -b/2$: then $\mathbb{E}\{w - Z\} = (1 - b)/2 \to 0$, also contradicting a positive uniform rate with finite fixed K. In fact $1 - |\cos(\pi b)|$ vanishes quadratically in $1 - b$, consistently with the rate in Lemma 5.1. □

6 An actual iid law with arbitrarily small moment resources

The obstruction can be realized by the actual conditional laws of a centered, variance-one iid marginal with a large central gap. It is not merely an optimizer of a moment relaxation.

For rational $\varepsilon, t > 0$, set

$$q = \frac{1 + \varepsilon - \varepsilon^2}{1 + t^2}, \quad p = 1 - q, \quad r = tq, \qquad 0 < \varepsilon < p < 1. \tag{11}$$

Let Y take the values $\varepsilon-1, \varepsilon, 1$ with respective probabilities $q\varepsilon, q(1-\varepsilon), p$, and put $X = (Y-p)/r$. Direct calculation gives

$$\mathbb{E}Y = p, \quad \mathrm{Var}(Y) = q\{p + \varepsilon(1-\varepsilon)\} = r^2.$$

Therefore X is exactly centered and standardized. The first two atoms are negative, and the third is positive. In the actual sign split, $\ell = 1/r$, $U_1 = 0$, and

$$U_0 = \begin{cases} \varepsilon - 1, & \text{probability } \varepsilon, \\ \varepsilon, & \text{probability } 1-\varepsilon. \end{cases}$$

Thus the actual conditional amplitude is $1-\varepsilon$, its span is one, the normalized central-gap increment is $c_{\mathrm{gap}} = 1-\varepsilon$, and $\lambda = nq\varepsilon(1-\varepsilon)$. In particular the central gap is wide even when the harmonic estimate fails completely.

For a fixed interior count k, the actual conditional sum has the law

$$V_k = (n-k)\varepsilon - \mathrm{Bin}(n-k, \varepsilon).$$

With $w = (n-k)\varepsilon$, its fractional-part gain is zero. For the four examples checked below, $n = 42$, $k = 17$ and $u = \eta = w/2$ give a legal phase $|u| \le 1/2$, $0 < \eta \le 1$, and $w = u + \eta$, exactly as in the marked-crossing argument. Its own-count variance is $(n-k)\varepsilon(1-\varepsilon) > 0$.

The physical defect is computed from the actual gaps, not from the two-point skeleton:

$$\rho = \mathbb{E}|X|^3, \quad \gamma = \mathbb{E}X^3, \quad J = \sum_{x_i < x_{i+1}} \left(\sum_{x_j \ge x_{i+1}} p_j x_j \right) (x_{i+1} - x_i)^2, \qquad D = (3+\sqrt{10})\rho - \gamma - 3J.$$

The exact checker uses rational moments and the directed enclosure

$$3.162277660168379331 < \sqrt{10} < 3.162277660168379332,$$

verified by squaring positive rationals. Its first law, with $\varepsilon = 1/100$, $t = 21/25$, is explicitly

x	$-198687/70693$	$-169661/212079$	$25/21$
$\mathbb{P}(X = x)$	$10099/1705600$	$999801/1705600$	$6957/17056.$

It has $|X| < 2.811$, $1.11975 < \rho < 1.11976$, and $0.67024 < D < 0.670241$. The four fixed examples satisfy:

ε	t	strict upper bound for D	strict upper bound for λ
10^{-2}	$21/25$	0.670241	0.246199
10^{-3}	$21/25$	0.070624	0.024625
10^{-6}	$21/25$	0.000738	0.000025
10^{-9}	$849/1000$	0.0000001	0.0000001

Every example satisfies its own recorded v6 moment-cell defect bound, weighted nD bound, and strict actual-gap criterion, as well as $D < 0.784$. This is a fresh exact comparison with the pinned table, not a replay of the Fourier calculation that originally produced that table.

The last example also meets every listed moment, probability and inner-gap bound of the retained response theorem's central row 19. Its sole failed listed domain bound is the support radius 1.84: its rare outer atom has magnitude greater than 2.875. Thus that theorem cannot be applied on the strength of its favorable moment bounds alone.

Corollary 6.1 (No moment-only amplitude implication)**.** *For every $\delta > 0$ there is an actual centered variance-one finite law supported in $[-96/25, 96/25]$, with positive conditional noise, $D < \delta$ and $\lambda < \delta$ at $n = 42$, whose negative conditional residual span is exactly one and whose fractional-part gain vanishes at a legal marked phase. The normalized central-gap increment can be arbitrarily close to one.*

Proof. At $\varepsilon = 0$ the positive-weight part of (11) is the standardized two-point law at $-t, 1/t$, with positive probability $p = t^2/(1+t^2)$. Its defect is

$$D_B = \frac{(6+2\sqrt{10})(p-p_*)^2}{\sqrt{p(1-p)}}, \qquad p_* = (4-\sqrt{10})/2.$$

This follows by substituting ρ_B, γ_B and $J_B = t + 1/t$. Choose positive rational t arbitrarily near $\sqrt{p_*/(1-p_*)}$. Then D_B is arbitrarily small. For this fixed t, let positive rational ε tend to zero. All moments are continuous. The extra outer-gap weight tends to zero, so the displayed finite gap formula gives $D \to D_B$ as well; $\lambda \to 0$ while the conditional span remains one. At the limiting value of t, the outer atom tends to $-(1+2t^2)/t$. Here $0.84 < t < 0.85$, so its magnitude is less than $2(0.85) + 1/0.84 < 3$. The support is therefore within $[-96/25, 96/25]$ in a neighborhood. The sign, phase and positive-noise conditions also hold for all sufficiently small ε. □

These laws are not asserted to satisfy a maximizing contact inequality, nonnegative fixed-target excess, or all new signed-acquisition outputs. They do not refute the sharp conjecture or the retained local theorem. They refute the proposed implication from small moment resources and an actual central gap alone to a positive fractional-part gain or a small conditional amplitude. Contact-based confinement or a quantitatively proved replacement for harmonic nonresonance remains an essential missing edge in that shortcut.

7 Evidence, scope and authorship

The exact source, proof object, fresh acceptance streams, input hashes and independent review reports accompany this note. The acceptance contract replays the two-quadratic ancestor, the support-radius gates, all selected signed boxes and the complete probability export. Independent reviews separately reconstruct the Fourier arguments, the moment and gap bounds, the fixed-target transport and the panel accounting. They are not all independent implementations of the same numerical program.

The named Fourier lemmas are proved above under their explicit moment and actual-event hypotheses. Their proof does not establish those hypotheses for an arbitrary law. Likewise the new coarse entry does not establish every subsequent confinement or count predicate. The original certificates and theorem targets remain unchanged. A complete finite base and ordered final exclusion must still be supplied before an unconditional sharp theorem is claimed.

AI systems were used to develop candidate arguments, write and revise the exposition, implement exact checks, and conduct separate adversarial reviews. The authors and readers must assess the displayed arguments and reproducible evidence; agreement between AI reviews is not a proof.